

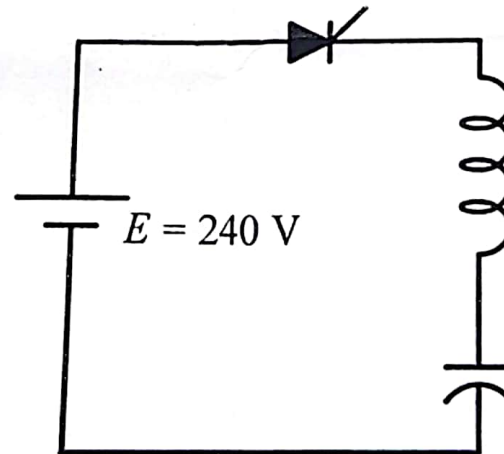
Chavla
200938

Time: 40 mins

CT-1

Marks: 15

- Q.1 Draw two transistors equivalent circuit of an SCR. Explain the SCR operation using two transistors model and gate triggering. 5
- Q.2 Assume the following circuit is initially relaxed and circuit is turned on at $t = 0$ s. Design the circuit parameters to turn off the thyristor at $t = 1$ ms. Also draw the circuit current and capacitor voltage waveform with proper levelling of peak amplitude and axes. 5



- Q.3 Explain the operation of class C or complementary commutation circuit with necessary diagram. 5

CT-2

Time: 30 mins

Marks: 15

- | | | |
|-----|---|---|
| Q.1 | Why string efficiency is always less than unity for series and parallel connected thyristors? Explain with suitable sketch. | 5 |
| Q.2 | Assume two thyristors T_1 and T_2 are connected in series, show the anode and string voltage distributions, if T_1 turn-on time is smaller than T_2 . | 5 |
| Q.3 | Explain the current equalization process by a centre-tapped inductor, with neat sketch, for parallel connected thyristors. | 5 |

CT-3

Time: 30 mins

Marks: 15

- | | | |
|-----|---|---|
| Q.1 | Draw a UJT relaxation oscillator circuit and explain its operation. | 5 |
| Q.2 | Which is the most sensitive mode for a Triac operation? Explain its operation. | 5 |
| Q.3 | What is a light activated SCR. Explain its working principle and mention its application. | 5 |

Course Title: Industrial & Power Electronics

Course No.: ECE 4101

Section B: Class Test 1 (Session: 2022-2023)

Date: 13-03-2024

Marks: 30

Time: 30 Minutes

- | | | | |
|-------|---|----|------|
| Q.(1) | Define stepper motor. List the advantages of stepper motors. | 08 | CO-2 |
| Q.(2) | Describe the operation of a variable reluctance type stepper motor with necessary diagrams. | 12 | CO-2 |
| Q.(3) | Explain the working principle of a bipolar stepper motor with necessary diagram. | 10 | CO-2 |

- Q.1) Define inverter. List the requirements of a practical inverter. (06)
- Q.2) Explain the working principle of a ^{single phase} full-bridge voltage source inverter with necessary diagrams. (16)
- Q.3) Draw the circuit diagram of a single phase half-wave converter. (08)

Time : 30 min
Start : 09:18
End : 09:48

CT 1, ECE 4103, Marks 30, Time 35m

1. In the inverter circuit, what is meant by $Z_{p,u}$ and $Z_{p,d}$? Derive the required ratio between $Z_{p,u}$ and $Z_{p,d}$ if an NMOS inverter is to be driven from another NMOS inverter. 12
2. For a MOS device, prove that transconductance is directly related by its geometry or difference between gate to source and threshold voltages (in saturation). 10
3. Determine the induced charge, drain to source current, gate/channel capacitance, and gate capacitance per unit area where oxide layer permittivity is 4, oxide layer thickness is $2\text{ }\mu\text{m}$, gate to source voltage 5 V, threshold voltage is 1 V, drain to source voltage is 5 V, channel length is $2\text{ }\mu\text{m}$, and channel width is $2.5\text{ }\mu\text{m}$, respectively. Assume that, this transistor is operated in non-saturation mode. 08

CT 2, ECE 4103, Marks 25, Time 35m

1. Draw the stick diagram and mask layout of the following expressions:
 $Y = \overline{A + B + C}$ use nMOS, and (ii) $Y = \overline{X \cdot Z}$ use CMOS 16
2. A particular layer of MOS circuit has a resistivity $\rho = 1 \text{ ohm cm}$. A section of this layer is $55 \text{ }\mu\text{m}$ long and $5 \text{ }\mu\text{m}$ wide 09
and has a thickness of $1 \text{ }\mu\text{m}$. Calculate the resistance from one end of this section to the other. What is the value of R_s ?

Charja

ECE 4103 CT1 Date: 24/03/2024 Marks: 10 Time: 20 min

1. Draw the stick diagram of an AND gate using 4:1 multiplexers according to CMOS technology. 05
2. Draw the stick diagram of the leaf cell of a shift register using nMOS technology. 05



জাহিদ

Active now



Q1. Stick diagram of nMOS adder.

Q2. Stick diagram of 3*3 barrel shifter



← জাহিদ replied to themselves

Q1. Stick diagram of nMOS adder.

Q2. Stick diagram of 3*3 barrel shifter



VLSI

2, sec-B

ECE 4104 Lab test (10) Date: 23/04/2024 Time 30 minutes

Question: See the board.

Common Instructions which you must follow

- i) File name: labtestyourID i.e. labtest200901
- ii) Use a, b as input variable names, Use op as output variable name
- iii) Use PIN_AC28 and PIN_AC27 as input pins for SW[1] and SW[2], respectively
- iv) Use PIN_G19 as output pin for LEDR[0]

CT-1(Closed-book), ECE 4123(A), Duration: 25 minutes, Marks: 25, Date: 04.02.2024, Time: 1:00 pm

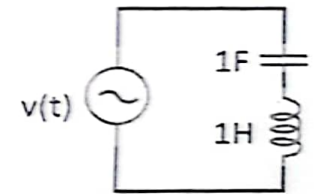
(Q1).(i) Draw the typical block diagram of a closed-loop control system which can describe a specific control problem including disturbance. Name each block and state that control problem. [3+3+2] (ii) For a control system engineer what is the main consideration of trade-off between choosing an open-loop and closed-loop design? [2].(Q2).Transfer function of an initially relaxed LTI system is given as $\frac{4}{s^2}$. Find the impulse response of the system and determine its output for an exponentially increasing input signal, e^t . [5+10].

CT-2(Open-book), ECE 4123(A), Duration: 25 minutes, Marks: 25, Date: 20.02.2024, Time: 11:00 am

(Q1). Mention three advantages and one limitation of using state-space modeling over classical approach in control systems design. [3×2+1=7]

(Q2). For the system shown in Fig.Q2, the controlled variable is the voltage across the inductor. (i) Obtain the state-space representation of the system, (ii) Solve for the state transition matrix, and (iii) Check whether the system is controllable/observable or both using Kalman test. [10+5+3=18]

Fig.Q2



QT(Closed-book), ECE 4123(A), Duration: 10 minutes, Marks: 05, Date: 04.03.2024, Time: 01:00 pm

(1).Open loop control systems are not affected by disturbances. (T/F) (2).The natural response of a control system does not depend on the input to the system. (T/F) (3).If the two real poles of a second order stable system are located on the same position of the s-plane, the system's step response will be *overdamped/undamped/critically damped/underdamped*. (4).For the nonminimum phase system, at least one zero lies on the *right-half/left-half/origin* of the s plane. (5).Type-1 system cannot follow the *ramp/parabolic/step* input.

- (Q1). Convert the block diagram of the control system (shown in Fig.Q1) to a signal-flow graph. (5)
- (Q2). Evaluate the steady state error of the control system (shown in Fig.Q1) for a standard ramp input. (15)
- (Q3). What information is contained in the specification $K_a=1000$? (5)

Sketch Bode plot for the transfer function

$$G(s) = \frac{1000}{s(1+0.1s)(1+0.001s)}$$

- 1) Gain cross over freq
- 2) Phase cross over freq
- 3) GM & PM
- 4) Stability of system

Marks 10 Time 25 min

1. Consider the transfer function

$$G(s)H(s) = \frac{60}{(s+1)(s+2)(s+5)}$$

11.08

Using Nyquist stability criterion determine whether the close loop system is stable or not.

200938
Course No: ECE 4125

CT1

Time: 25 minutes

Full Marks: 15

1. What is division point? What is the significance of it in medium resistance measurement? 03
2. What is ground and short circuit fault? Derive the equation of maximum bridge sensitivity for Wheatstone bridge. 06
3. A short circuit fault is located by Varley loop test. The ratio arms are set at $P = 5\Omega$, and $Q = 10\Omega$ and the values of variable resistance S are 16Ω for position 1 of switch K and 7Ω for position 2. The sound and faulty cables are identical and have a resistance of $0.4\Omega/\text{km}$. Determine the length of each cable and the distance of the fault from the test end. 06

Course No: ECE 4125

CT2

Time: 25 minutes

Full Marks: 15

1. Why Maxwell's inductance capacitance bridge is suited for a limited range of Q-coils? 5
Which bridge is useful for higher values of Q and why? 04
2. Derive the balance equation and quality factor of Hay's bridge. 06
3. In a Maxwell's inductance capacitance bridge $R_1 = 240\Omega$, $R_3 = 600\Omega$, $R_4 = 1000\Omega$, $C_4 = 0.5\mu F$ and frequency = 1000Hz. Determine R_2 , L_1 and quality factor. 05

Course No: ECE 4125

CT3

Time: 30 minutes

Full Marks: 15

1. What is inverse transducer? Briefly discuss the working mechanism of an active transducer. 04
2. Define resistive transducer. Justify how an inductive transducer is used for the measurement of displacement. 05
3. A parallel plate capacitive transducer has a plate area ($40\text{mm} \times 40\text{mm}$) and capacitance 28.3 pF . Determine the plate spacing and displacement that causes the capacitance to change by 5 pF . Also, determine the transducer sensitivity. 06

Course No: ECE 4125

CT2

Time: 25 minutes

Full Marks: 15

- C.T-4
1. What is quantisation error? For a D/A converter, justify that the analog output voltage is proportional to the digital input. 05
 2. Explain the operation of a $3\frac{1}{2}$ -digit display counter. 05
 3. An analog transducer with a $0 - 10\text{ V}$ input is able to distinguish a change of 10 mV in its input signal. Determine its resolution, the number of bits of an A/D converter so that the digital output has almost the same resolution as the transducer, the quantization error and the number of decision levels. 05

CT 2, ECE 4107, Marks-30, Time-30 minutes

- A 10000-line exchange has 400 GS and 450 FS uniformly distributed. How many simultaneous calls can be put through the exchange? Also, determine various design parameters. 18
- “Numbering plan in a telephone network must be independent of call routing” Why? Explain. 12
- this can happen. 12
- Draw the schematic of a switch system and explain its function. 10
- Prove mathematically that carbon microphone acts as an amplitude modulator.

CT 1, ECE 4107, Marks 30, Time 35m

Draw the pulse dialing waveform for the following number : 4134
In Strowger exchanges, a call may be blocked even though an appropriate path through the switch exists. Explain how this can happen.

4
4

Draw the schematic of a switch system and explain its function.

Prove mathematically that carbon microphone acts as an amplitude modulator.

12
10

Schematic of an

1. Draw the X.400 message handling system.
2. Suppose an arrival rate of $5/s$ and service rate of $4/s$. Calculate the mean waiting time of a 5-service ISDN and a 5-service segregated system.